

TrainAudit

OFFLINE — mine once per framework

1 Invariant Miner

LLM-instantiated candidates, untrusted until verified



Instantiate

Scope

fix training phase + parameter class

Ground

evidence $\mathcal{E}^{\text{LLM}}$ from framework source & docs

verify (decisive)

Construct

counterexample attack ≥ 2 CEs; one CE holds $\Rightarrow$ reject outright

Accept

no CE holds $\wedge$ Conf(c) ≥ 0.8 ($\theta = 0.8$)

reject — next iteration

five-stage FSM

S_1

gap analysis

S_2

evidence retrieval

S_3

synthesis+attack

S_4

persistence

S_5

reporting

decisive counterexample gate

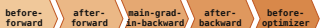
healthy-run replay (verification only)

Conf(c) ≥ 0.8

ONLINE — every training job · fully deterministic · no LLM

2 Data Collector

no user code changes · adapter 30-150 LoC



+ auxiliary taps (ckpt · all_reduce · snapshot) = 8 hookpoints

GPU-side scalar reductions (checksums only when required)

rank / step / stage unrolled into columns written to DuckDB trace (two-step window)

captured record (S0):

cksum param_name rank step stage

DuckDB trace DB

two-step sliding window

3 Verifier

deterministic SQL checking — no LLM

at job start

extract topology : topology-aware pruning · lineage: replicated vs. sharded

per step, per rule

WHERE $R_{\text{step}} \wedge R_{\text{expected}}$
GROUP BY param, step
HAVING R_{cksum} violated

empty result = holds · non-empty IS the violation set

coarse → fine: checksum → module → param → bitwise



Verified Constraint Library P
guarded relational constraints

P3 · cross-rank replication
(SwitchMLP router sync)

R_{type} : TP > 1
 R_{expected} : tpi = False
(not tensor-parallel sharded)
 R_{schema} : replicas share checksum

compiled to parameterized SQL at job start
accepted rules determine trace schema (S0-S6)
✓ 0 false positives over 764 clean evals

violation report

param	rank	cksum
router.weight	0	cksum ₁
router.weight	1	cksum ₀

x divergent checksums

- ✓ violating constraint
- ✓ affected parameter
- ✓ divergent ranks
- ✓ first violating step

offline inputs

Pattern Catalog

template P3:
pi_topo: TP > 1
pi_precond: tpi=False
pi_schema: share cksum
(cross-rank replicas)
from 392 silent errors

35 templates · 13 fault classes

Framework source

Megatron-LM
DeepSpeed
OLMo / OLMo-core
silent-vs-loud filter
no raise / assert

once per framework

online output

Violation Report

violating constraint: P3
affected parameter:
router.weight
divergent ranks: 0 vs 1
(SwitchMLP router sync)
non-empty SQL result

empty result = holds

non-empty = violation set

Legend

offline mining

online checking

reject / violation LLM only inside offline Miner · online path: no LLM